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EE-340 Assignment -4

Ans 1 We have,

$$\text{at generator, } \Gamma_g = \frac{Z_g - Z_0}{Z_g + Z_0} = \frac{25 - 50}{25 + 50} = -1/3$$

$$\text{at load, } \Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{100 - 50}{100 + 50} = 1/3$$

$$\text{and } V_1^+ = \frac{Z_0 V_g}{Z_g + Z_0} = \frac{50 \times 10}{25 + 50} = 6.67 \text{ V}$$

$$V_1^- = \Gamma_L V_1^+ = 2.22 \text{ V}$$

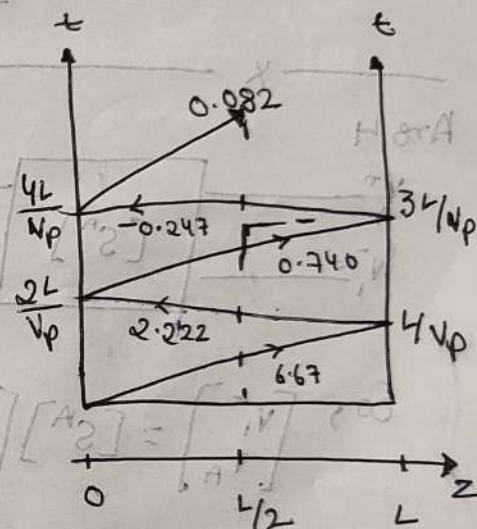
$$V_2^+ = \Gamma_g V_1^- = -0.74 \text{ V}$$

$$V_2^- = \Gamma_L V_2^+ = -0.247 \text{ V}$$

$$V_3^+ = \Gamma_g V_2^- = 0.082 \text{ V}$$

$$\text{At } z = \frac{L}{2} \text{ and } t = \frac{3L}{V_p}$$

$$V_T = V_1^+ + V_1^- + V_2^+ = 8.149 \text{ V}$$



Ans 2 Normalized $Z_L = \frac{Z_L}{Z_0} = 1.2 + j1$

From the Smith Chart (at end)

(a) $SWR = 2.45$

(d) $Z_{in} = Z_{in} Z_0 = 24.5 + j20.5 \Omega$

(b) $\Gamma = 0.42 \angle 54^\circ$

(c) $L_{min} = 0.325 \lambda$

(c) $Y_L = \frac{Y_L}{Z_0} = \frac{0.49 - j0.41}{50} = 0.0098 - j0.0082$

(f) $L_{max} = 0.075 \lambda$

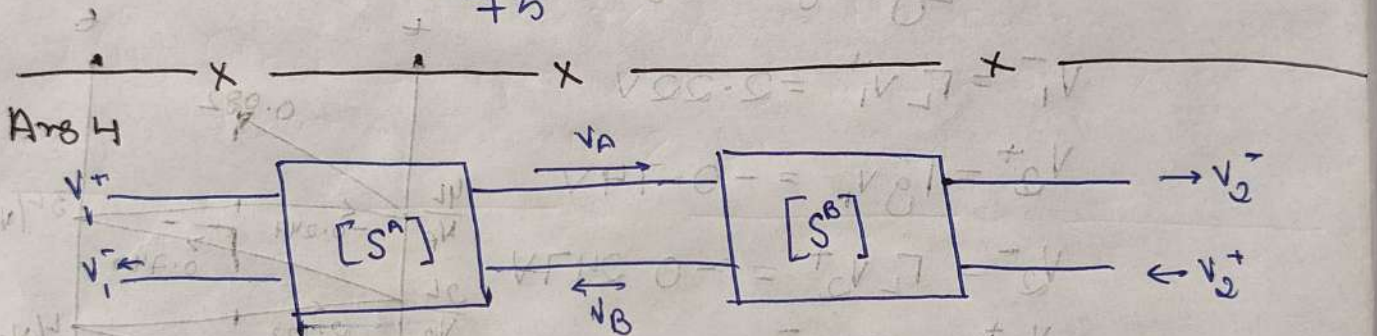
Ans 3 (a) $Z_{in} = \frac{Z_{in}}{Z_0} = 0 \Rightarrow l = 0 \text{ or } 0.5 \lambda$

(b) $Z_{in} = \frac{Z_{in}}{Z_0} = \infty \Rightarrow l = 0.25 \lambda$

(c) $Z_{in} = \frac{j75}{75} = j1 \Rightarrow l = 0.125 \lambda$

(d) $Z_{in} = \frac{-j50}{75} = -0.667j \Rightarrow l = 0.406 \lambda$

(e) $Z_{in} = \frac{10j}{75} = 0.133j \Rightarrow l = 0.021 \lambda$



So, $\begin{bmatrix} V_1^+ \\ V_B \end{bmatrix} = [S^A] \begin{bmatrix} V_1^+ \\ V_2^- \end{bmatrix} \quad \text{--- (i)}$

$\begin{bmatrix} V_B \\ V_2^- \end{bmatrix} = [S^B] \begin{bmatrix} V_A \\ V_2^+ \end{bmatrix} \quad \text{--- (ii)}$

And, $\begin{bmatrix} V_1^- \\ V_2^- \end{bmatrix} = [S] \begin{bmatrix} V_1^+ \\ V_2^+ \end{bmatrix} \Rightarrow S_{21} = \frac{V_2^-}{V_1^+} \bigg|_{V_2^+ = 0}$

When $V_2^+ = 0$, from (ii)

$V_B = S_{11}^B V_A \quad \text{--- (iii)}$

$V_2^- = S_{21}^B V_A \quad \text{--- (iv)}$

From (i)

$V_A = S_{21}^A V_1^+ + S_{22}^A V_B = S_{21}^A V_1^+ + S_{22}^A S_{11}^B V_A \quad [\text{Using (iii)}]$

$\frac{V_A}{S_{21}^A} = V_1^+ + S_{22}^A S_{11}^B V_A \quad [\text{Using (iv)}]$

$$= V_2^- \left(\frac{1 - S_{22}^A S_{11}^B}{S_{21}^B} \right) = S_{21}^A V_1^+$$

$$\text{So, } S_{21} = \frac{V_2^-}{V_1^+} \bigg|_{V_2^+ = 0} = \frac{S_{21}^A S_{21}^B}{1 - S_{22}^A S_{11}^B}$$

Ans 5 $Z_{L, \text{Norm}} = \frac{Z_L}{Z_0} = \frac{100 + j80}{75} = 1.33 + j1.067$

From Smith chart (At end)

$$y_1 = 1 + 0.95j$$

$$y_2 = 1 - 0.95j$$

Distance d from load to stub

$$d_1 = (0.66 - 0.434)\lambda = 0.226\lambda$$

$$d_2 = (0.84 - 0.434)\lambda = 0.406\lambda$$

Req length of short-circuited stubs

$$l_1 = (0.379 - 0.25)\lambda = 0.129\lambda$$

$$l_2 = (0.621 - 0.25)\lambda = 0.371\lambda$$

Two design solns --

(i) Place a shorted stub of length 0.129λ at distance 0.226λ from the load.

(ii) Place a shorted stub at 0.371λ at distance of 0.406λ from the load.

Ans 6 $Z_{L, \text{Norm}} = \frac{Z_L}{Z_0} = \frac{90 + j60}{75} = 1.2 + j0.8j$

From Smith chart (at end)

$$y_1 = 1 + 0.75j$$

$$y_2 = 1 - 0.75j$$

Distance d from load to stub,

$$d_1 = (0.653 - 0.423)\lambda = 0.23\lambda$$

$$d_2 = (0.847 - 0.423)\lambda = 0.424\lambda$$

Req length of open-circuited stubs

$$l_1 = 0.398\lambda$$

$$l_2 = 0.102\lambda$$

∴ Two Design Solns -:

(i) Place an open stub of length 0.398λ at a distance of 0.23λ from the load.

(ii) Place an open stub of length 0.102λ at a distance of 0.424λ from load.

$$R_{255.0} = R_{in}(150.0 - 10.0) = 1.0$$

$$R_{100.0} = R_{in}(150.0 - 130.0) = 1.0$$

solve between - to find the length of

$$R_{251.0} = R_{in}(25.0 - 675.0) = 1.0$$

$$R_{175.0} = R_{in}(25.0 - 150.0) = 1.0$$

to $R_{251.0}$ - to find the length of stub at distance 0.23λ from the load.

book at more $R_{255.0}$ distance

to $R_{175.0}$ to find the length of stub at distance 0.424λ from the load.

book at more $R_{100.0}$ of

$$28.0 + 0.1 = \frac{201 + 0.8}{2.5} = \frac{2.5}{0.5} = 5.0$$

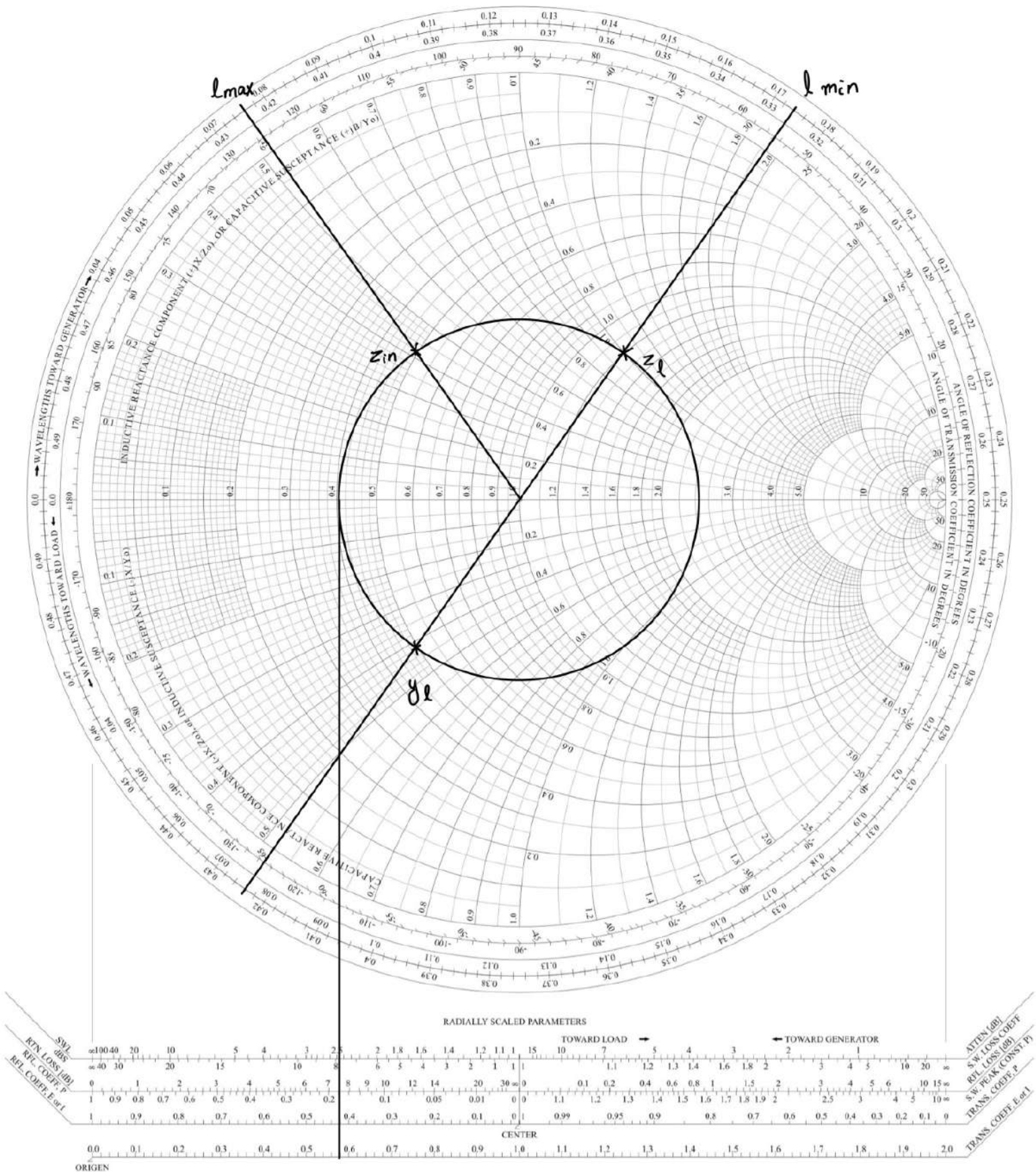
(bring to) back thing more

$$28.0 + 1 = 29$$

$$28.0 - 1 = 27$$

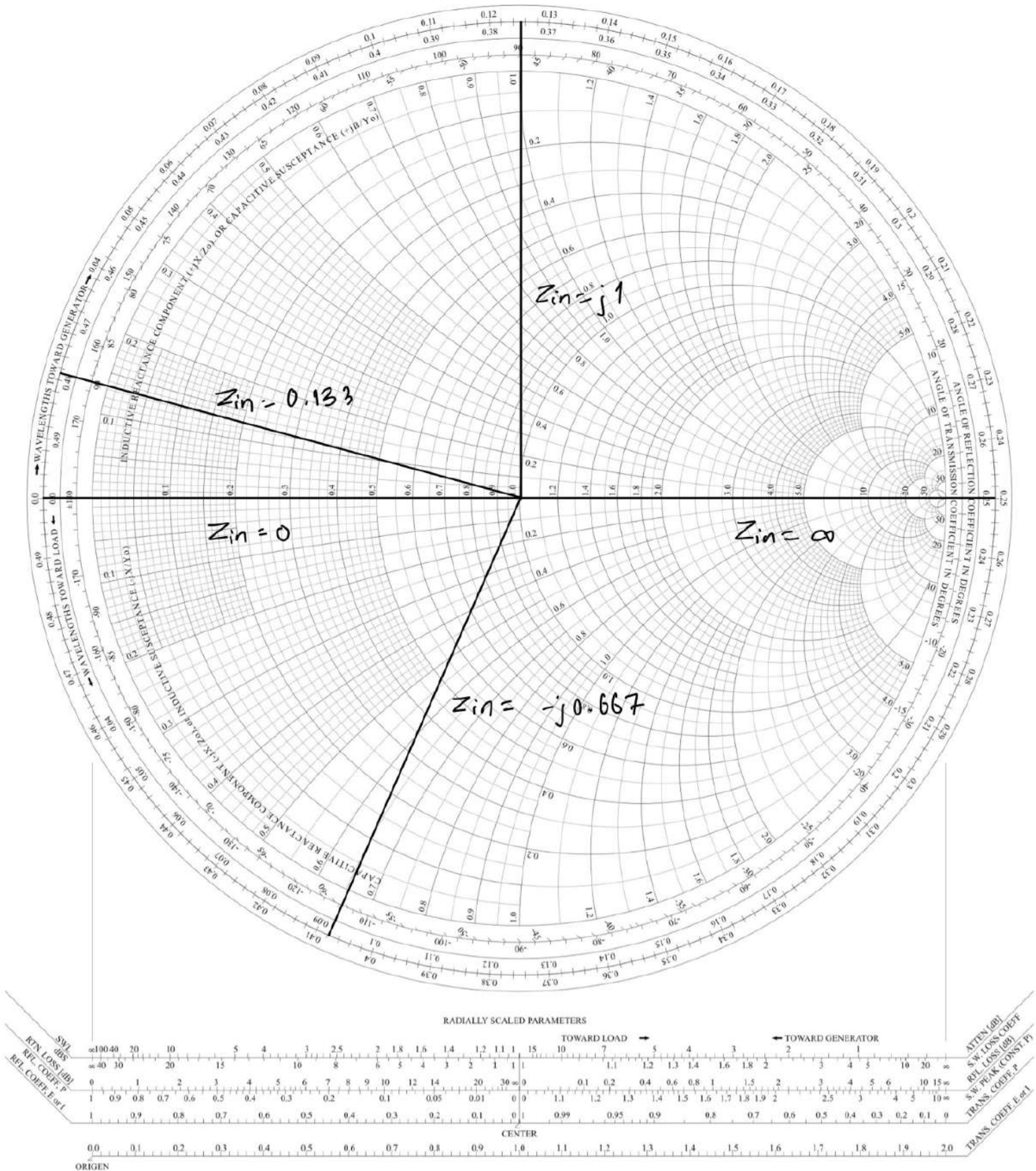
Question 2

SMITH CHART



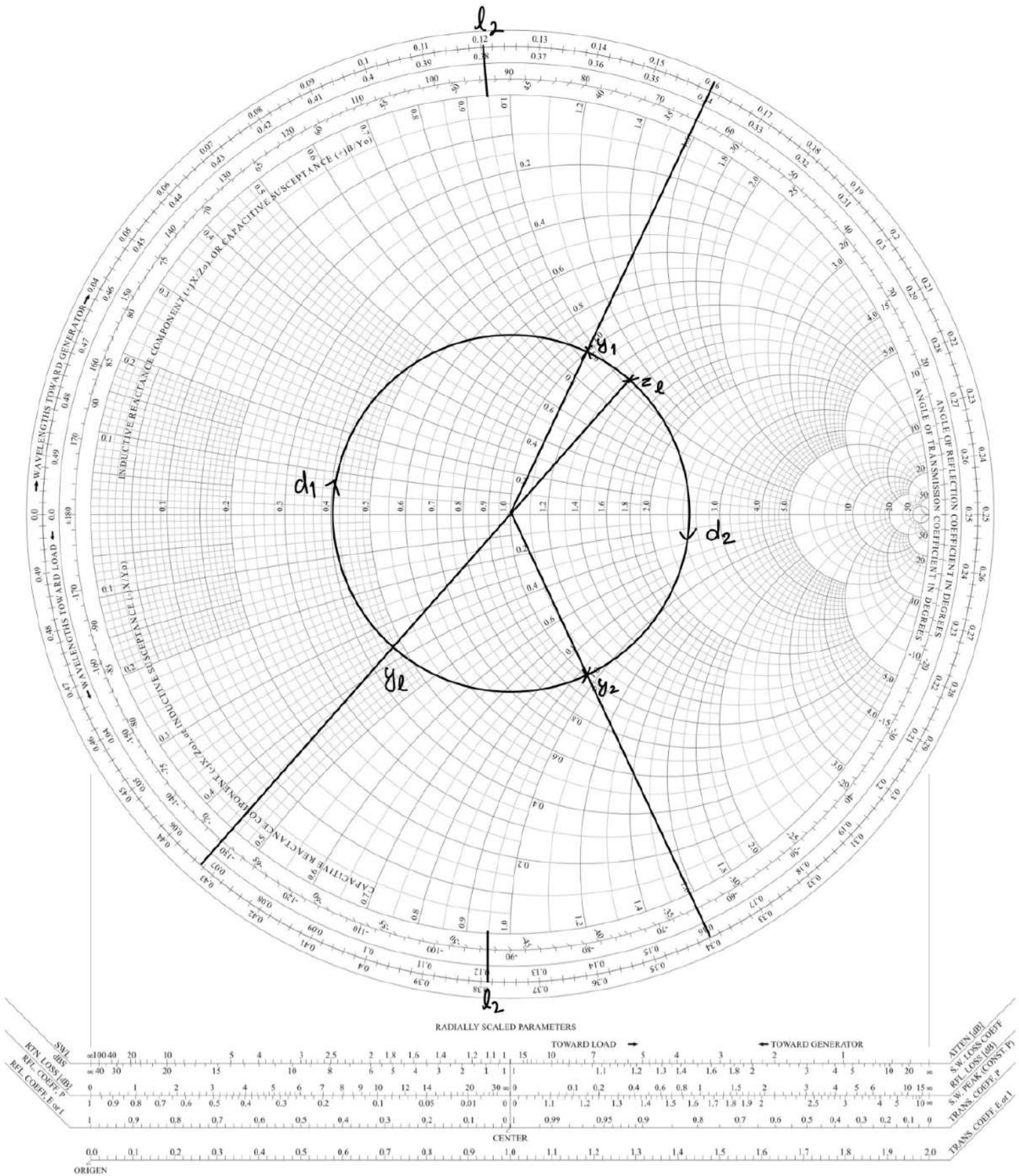
Question 3

SMITH CHART



Question 5

SMITH CHART



Question 6

SMITH CHART

